

M003 Week 1 — English Worksheet

Topic: First Steps with Coordinates (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you draw on a flat **wall**. Every spot on the wall gets a name made of **two numbers: (x, y)**. x tells you how far **across**, y tells you how far **up**. Counting starts at **0**.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
place gold block at (3, 0)
place gold block at (3, 5)
```

The two gold blocks share the same x. Are they close together or far apart? Which number changed?


```
place red block at (0, 0)
place red block at (4, 0)
place red block at (0, 4)
```

Three red blocks. Which one sits at the corner where counting starts? Which one is higher up the wall?


```
set y to 2
place gold block at (0, y)
set y to 6
place gold block at (0, y)
```

The variable y changes from 2 to 6. Where does each gold block land? Do they share the same x ?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The block should land at **(2, 5)** — 2 across on x, 5 up on y. Right now it lands somewhere else.

```
place gold block at (5, 2)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The block should sit at the **bottom corner** of the wall. The coder forgot where counting starts.

```
place gold block at (1, 1)
```

Hint: counting starts at **0**, not 1.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The block should sit **5 up** from the ground at $x = 0$ — that is **(0, 5)**. Right now it goes the wrong way and drops below the wall.

```
place gold block at (0, -5)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Place a gold block at **(0, 0)** so you can see where counting starts. Then pick three more coordinates, place a different block at each one, and write down the coordinates you used. When you finish, come back here.

Send a photo or video of your blocks, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

The block at **(0, 0)** shows ...

I placed a block at (... , ...) because ...

The y number means ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you show your blocks on the wall. Talk through it like you are teaching someone who has never seen coordinates. Try to use these words: **coordinate, position, x, y, zero.**

1. Show the gold block at $(0, 0)$ and say why it is there.
2. Point at one of your blocks and say its coordinate out loud.
3. Pick two blocks and say which one is **higher** and how you know from the y number.
4. Say in your own words what **x** and **y** mean.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.